Case Math and Estimation Drills

Consulting Technical Curriculum companion to C03. 12 drills with worked answers.

Cover the answer, solve out loud, then check. Round aggressively, state every assumption, and finish each one with a sanity check. There is no single right number on sizing drills, only a defensible method.

Drill 1

Estimate the number of coffee shops in a city of 1M people.

APPROACH

Bottom-up from demand. ~1M people, assume ~40% buy coffee out, ~2 cups/week. That is ~800K cups/week. A shop serves perhaps ~700 cups/day, ~5,000/week.

ANSWER

800K / 5,000 is about 160 shops. Sanity check: one shop per ~6,000 people feels reasonable for a mid-size city.

Drill 2

A SaaS app has $3M fixed cost. Price $150/yr, variable cost $30/user. Breakeven users?

APPROACH

Contribution margin per user is $150 - $30 = $120. Breakeven = fixed / margin.

ANSWER

$3M / $120 = 25,000 users. Each user past that adds $120 profit.

Drill 3

A product's profit fell while revenue was flat. Where do you look?

APPROACH

Profit = Revenue - Cost. Revenue flat, profit down, so cost rose. Split cost into fixed and variable, then per-unit.

ANSWER

Find the cost line that rose. With volume flat, a higher total cost means higher cost per unit, pointing to input prices or waste.

Drill 4

Revenue grew from $80M to $100M in two years. Annual growth rate?

APPROACH

Total growth is 25% over two years. The compound annual rate solves $(1+r)^2 = 1.25$.

ANSWER

r is about 12% per year. The simple average of 12.5% per year overstates slightly because of compounding.

Drill 5

An income statement shows Revenue $200, COGS $120, Operating expense $40, Tax $8. Find the three margins.

APPROACH

Gross = (Rev - COGS)/Rev. Operating = (Rev - COGS - OpEx)/Rev. Net = (Operating profit - Tax)/Rev.

ANSWER

Gross 40%, Operating 20%, Net 16%. Know which costs each margin includes.

Drill 6

Quick: what is 17% of 250?

APPROACH

Percentages are reversible. 17% of 250 = 250% of 17 = 2.5 x 17.

ANSWER

2.5 x 17 = 42.5.

Drill 7

A factory has 3 stations running at 90, 50, and 70 units/hour. Line output?

APPROACH

Output equals the slowest station, the bottleneck.

ANSWER

50 units/hour, set by station 2. Speeding up the others does nothing until station 2 is relieved.

Drill 8

A machine costs $400K and saves $120K/year. Simple payback?

APPROACH

Payback = cost / annual saving.

ANSWER

$400K / $120K is about 3.3 years. Note this ignores the time value of money, which a fuller analysis would add.

Drill 9

Estimate annual umbrella sales in a rainy city of 2M adults.

APPROACH

Assume each adult owns ~2 umbrellas and replaces one every ~2 years, so ~1 purchase per adult per year on average across the population, adjusted down for those who rarely buy.

ANSWER

Roughly 1M to 2M units a year. State the replacement-cycle assumption clearly; that is what is being graded.

Drill 10

A price cut from $50 to $45 raises volume from 100K to 130K units. Variable cost is $30. Better or worse?

APPROACH

Compare contribution. Old: ($50-$30) x 100K = $2.0M. New: ($45-$30) x 130K = $1.95M.

ANSWER

Slightly worse. The 30% volume gain did not offset the margin drop. Always compare total contribution, not revenue.

Drill 11

Estimate the US market for running shoes in units per year.

APPROACH

Top-down. ~330M people, assume ~25% are active runners or buyers, each buys ~1 pair every ~1.5 years.

ANSWER

~80M buyers x ~0.7 pairs/yr is about 55M pairs a year. Sanity check against population; one pair per ~6 people per year is plausible.

Drill 12

A store's sales are down 15%. Foot traffic is flat. What happened?

APPROACH

Sales = traffic x conversion x average ticket. Traffic flat, so conversion or ticket fell.

ANSWER

Either fewer visitors bought (conversion) or buyers spent less per visit (ticket). Ask for conversion and average basket to isolate which.